AP CYBERSECURITY | INVESTIGATION LAB - STUDENT

LAB 1

The School Account Breach

Unit 1: Introduction to Security | 45 to 60 minutes

Your role

You are a junior security analyst for a fictional school district. A teacher reports being locked out of the grade portal, and a counselor notices grades that do not match the teacher's records. Your job is to work out **what happened, how, and what to do next**, using only the evidence files.

What you will do

- Read a phishing email and name its warning signs
- Build a timeline from a portal log and spot the anomalous login
- Sort the damage into confidentiality, integrity, and availability
- Name the threat, vulnerability, exploit, impact, and risk
- Recommend containment, recovery, and prevention steps

EXAM LENS
Every lab follows the same four stages: **Learn the concept, Investigate the evidence, Make a security decision, then Answer AP-style questions.** Commands show you what happened. The skill is deciding what the evidence *proves*, what it *does not prove*, and what to do next.

COMMON TRAP
This is a **fictional, controlled** investigation. Never scan, test, or access real school systems, other people's accounts, or any device you do not own or have written permission to use.

Original practice material written in the style of the AP Cybersecurity exam. Not reproduced from College Board.

Mission Briefing

On Wednesday morning, Mr. Rivera (portal user `mrivera`) could not log in to the school grade portal. A counselor also noticed that five students in his Period 3 Algebra class suddenly have much higher grades. You have three files. Work through the stages in order. You may use a terminal (for example `grep mrivera portal.log`) or simply read the files.

Evidence files

File	What it is
portal.log	Portal and web-filter log, Oct 5 to Oct 7 (user, time, source address, action)
phish_email.txt	A copy of an email Mr. Rivera received on Oct 6
grade_changes.csv	The portal's record of recent grade changes

Normal for this school

- Staff computers use addresses that begin with `10.8.` (the school network).
- Teachers normally log in between about 7:30 and 8:30 AM.
- The portal currently uses only a username and password (no multifactor authentication).

STAGE 1 Learn

Be able to explain these before you touch the evidence

- ■ I can name the three parts of the CIA triad and give one example of each.
- ■ I can explain the difference between a threat, a vulnerability, an exploit, and a risk.
- ■ I can name the phases of an attack: reconnaissance, initial access, persistence, lateral movement, evading detection, taking action.
- ■ I can list two warning signs of a phishing email and two defenses against it.

STAGE 2 Investigate

Collect facts. Record what you observe, not what you assume.

1. Open `phish_email.txt`. List **three warning signs** that the message may not be legitimate.

2. Filter the log for Mr. Rivera (`grep mrivera portal.log`). What **time of day** and **source address** does he normally log in from on Oct 5 and Oct 6?

3. Find the one login that does **not** fit his normal pattern. Record the date, time, and source address, and explain **two** reasons it is suspicious.

4. Look at the two `WEBFILTER` lines from Oct 6. What do they suggest Mr. Rivera did, and **how sure** can you be? Use the words 'confirmed' or 'likely'.

```
2026-10-06 15:20:44 WEBFILTER user=mrivera ip=10.8.4.22 visited hancock-mail-verify.example.net/login
2026-10-06 15:21:31 WEBFILTER user=mrivera ip=10.8.4.22 form_post hancock-mail-verify.example.net/login
```

5. Build a **timeline** of six key events from Oct 6 to Oct 7 (time and what happened).

6. For each event below, name the **CIA property** most directly affected and explain in one sentence.

Event	Log evidence
A. Gradebook downloaded by the adversary	EXPORT gradebook=Period3_Algebra rows=31
B. Student grades altered	five GRADE_CHANGE lines; grade_changes.csv
C. Mr. Rivera cannot get in	PASSWORD_CHANGE at 02:25; LOGIN FAIL at 07:45 and 07:46

7. Match entries in `portal.log` to attack phases. Which entry shows **initial access**? Which shows **persistence**? Which show **taking action**?

8. Complete the table with the **threat**, **vulnerability**, **exploit**, **impact**, and a **risk rating** (low, moderate, or high) with a reason.

9. Use `grade_changes.csv`. How many grades were changed, and for which student IDs? (Try `cut -d, -f1 grade_changes.csv` or `grep -c GRADE_CHANGE portal.log`.)

STAGE 3 Make a Security Decision

Turn evidence into a defensible recommendation

What can you confirm from the evidence? (at least three facts)

What is still uncertain? (at least two things the evidence does not prove)

What do you recommend? Give one immediate containment step, one recovery step, and two prevention controls.

STAGE 4 AP-Style Questions

Transfer what you learned to a new situation

1. In the breach, an adversary changed five students' grades without authorization.

Which CIA property was most directly affected by the grade changes?

- **A.** Confidentiality
- **B.** Integrity
- **C.** Availability
- **D.** Nonrepudiation

2. An analyst reviews the following portal entries.

```
02:14:07 LOGIN OK user=mrivera ip=198.51.100.77
02:21:45 GRADE_CHANGE student=S1043 from=F to=A
02:26:02 RECOVERY_EMAIL_CHANGE user=mrivera new=z9k.backup@mailbox.example
02:31:50 LOGOUT user=mrivera
```

Which entry best demonstrates persistence?

- **A.** 02:14:07
- **B.** 02:21:45
- **C.** 02:26:02
- **D.** 02:31:50

3. The adversary used a valid username and password obtained through phishing.

Which control would most likely have stopped the login even though the password was known?

- **A.** Locking the server room door
- **B.** Requiring password changes every 90 days
- **C.** A privacy screen on the teacher's monitor
- **D.** Requiring multifactor authentication for portal logins

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4. A security tool compares each login to a user's usual time and address and raises an alert for the 2:14 AM login from an unfamiliar address.

Which detection method does this describe?

A. Anomaly-based detection
B. Signature-based detection
C. A cryptographic hash check
D. A honeypot

Evidence Printout

Printed copies of the evidence files, for students without a terminal. The files themselves are in the evidence folder.

portal.log

```
2026-10-05 07:41:02 LOGIN OK user=mrivera ip=10.8.4.22
2026-10-05 07:55:40 LOGIN OK user=kosei ip=10.8.4.31
2026-10-05 08:03:12 LOGIN OK user=dthomas ip=10.8.5.14
2026-10-05 14:20:18 VIEW gradebook=Period1_Geometry user=kosei ip=10.8.4.31
2026-10-06 07:44:19 LOGIN OK user=mrivera ip=10.8.4.22
2026-10-06 07:58:03 LOGIN OK user=kosei ip=10.8.4.31
2026-10-06 08:09:47 LOGIN OK user=dthomas ip=10.8.5.14
2026-10-06 15:12:30 MAIL_DELIVERED user=mrivera subject="Mailbox storage almost full - verify now"
2026-10-06 15:20:44 WEBFILTER user=mrivera ip=10.8.4.22 visited hancock-mail-verify.example.net/login
    (category: uncategorized)
2026-10-06 15:21:31 WEBFILTER user=mrivera ip=10.8.4.22 form_post hancock-mail-verify.example.net/login
2026-10-07 02:14:07 LOGIN OK user=mrivera ip=198.51.100.77
2026-10-07 02:15:30 VIEW gradebook=Period3_Algebra user=mrivera ip=198.51.100.77
2026-10-07 02:16:12 EXPORT gradebook=Period3_Algebra rows=31 user=mrivera ip=198.51.100.77
2026-10-07 02:21:45 GRADE_CHANGE student=S1043 from=F to=A user=mrivera ip=198.51.100.77
2026-10-07 02:22:02 GRADE_CHANGE student=S1017 from=D to=A user=mrivera ip=198.51.100.77
2026-10-07 02:22:19 GRADE_CHANGE student=S1088 from=F to=B user=mrivera ip=198.51.100.77
2026-10-07 02:22:38 GRADE_CHANGE student=S1051 from=D to=A user=mrivera ip=198.51.100.77
2026-10-07 02:23:04 GRADE_CHANGE student=S1120 from=F to=A user=mrivera ip=198.51.100.77
2026-10-07 02:25:10 PASSWORD_CHANGE user=mrivera ip=198.51.100.77
2026-10-07 02:26:02 RECOVERY_EMAIL_CHANGE user=mrivera new=z9k.backup@mailbox.example ip=198.51.100.77
2026-10-07 02:31:50 LOGOUT user=mrivera ip=198.51.100.77
2026-10-07 07:45:21 LOGIN FAIL user=mrivera ip=10.8.4.22
2026-10-07 07:46:00 LOGIN FAIL user=mrivera ip=10.8.4.22
2026-10-07 07:48:30 HELPDESK_TICKET id=4471 user=mrivera note="locked out of portal"
2026-10-07 07:59:12 LOGIN OK user=kosei ip=10.8.4.31
2026-10-07 08:10:33 LOGIN OK user=dthomas ip=10.8.5.14
```

phish_email.txt

```
From: IT Helpdesk <helpdesk@hancock-mail-verify.example.net>
To: mrivera@school.example
Date: Tue, 06 Oct 2026 15:12:30 -0500
Subject: Mailbox storage almost full - verify now

Dear Teacher,

Our records show your mailbox is 98% full. To avoid losing incoming mail,
you must verify your account within 24 hours or it will be suspended per
district IT policy.

Verify here: https://hancock-mail-verify.example.net/login

Thank you,
IT Helpdesk Team
```

grade_changes.csv

```
student_id,course,old_grade,new_grade,changed_at,changed_by
S1043,Period3_Algebra,F,A,2026-10-07 02:21:45,mrivera
S1017,Period3_Algebra,D,A,2026-10-07 02:22:02,mrivera
S1088,Period3_Algebra,F,B,2026-10-07 02:22:19,mrivera
S1051,Period3_Algebra,D,A,2026-10-07 02:22:38,mrivera
S1120,Period3_Algebra,F,A,2026-10-07 02:23:04,mrivera
```

KEY IDEA

Before you turn this in: every answer should name the evidence it comes from (a file name, a line, a time, or a command result). If you are not sure something is proven, write 'not confirmed' and say what evidence would confirm it.